

● EASY

● ADVANCED MATH

Nonlinear equations in one variable and systems of equations in two variables

03

10 questions · ~15 min

Q1

ADVANCED MATH

$$y = 76$$
$$y = x^2 - 5$$

The graphs of the given equations in the xy -plane intersect at the point (x, y) . What is a possible value of x ?

A. $-\frac{76}{5}$

B. -9

C. 5

D. 76

$$x + 2x - 5x + 9 = 0$$

What is a positive solution to the given equation?

- A. 3
- B. 4
- C. 5
- D. 18

The total revenue from sales of a product can be calculated using the formula $T = PQ$, where T is the total revenue, P is the price of the product, and Q is the quantity of the product sold. Which of the following equations gives the quantity of product sold in terms of P and T ?

A. $Q = \frac{P}{T}$

B. $Q = \frac{T}{P}$

C. $Q = PT$

D. $Q = T - P$

$$x + 7 = 10$$

$$x + 7^2 = y$$

Which ordered pair x, y is a solution to the given system of equations?

A. 3, 100

B. 3, 3

C. 3, 10

D. 3, 70

$$6r = 7s + t$$

The given equation relates the variables r , s , and t . Which equation correctly expresses s in terms of r and t ?

A. $s = 42r - t$

B. $s = 76r - t$

C. $s = \frac{6}{7}r - t$

D. $s = \frac{6r - t}{7}$

$$P = \frac{W}{t}$$

The power P produced by a machine is represented by the equation above, where W is the work performed during an amount of time t . Which of the following correctly expresses W in terms of P and t ?

- A. $W = Pt$
- B. $W = \frac{P}{t}$
- C. $W = \frac{t}{P}$
- D. $W = P + t$

$$8j = k + 15m$$

The given equation relates the distinct positive numbers j , k , and m . Which equation correctly expresses j in terms of k and m ?

A. $j = \frac{k}{8} + 15m$

B. $j = k + \frac{15m}{8}$

C. $j = 8(k + 15m)$

D. $j = \frac{k + 15m}{8}$

$$c - 7 = 25p + k$$

The given equation relates the positive numbers c , p , and k . Which equation correctly expresses c in terms of p and k ?

A. $c = 25p + k + 7$

B. $c = 25p + k - 7$

C. $c = 725p + k$

D. $c = \frac{25p + k}{7}$

$$p + 34 = q + r$$

The given equation relates the variables p , q , and r . Which equation correctly expresses p in terms of q and r ?

- A. $p = q + r + 34$
- B. $p = q + r - 34$
- C. $p = -q - r + 34$
- D. $p = -q - r - 34$

$$x + 45 = 48$$

What is the positive solution to the given equation?

- A. 3
- B. 48
- C. 93
- D. 96